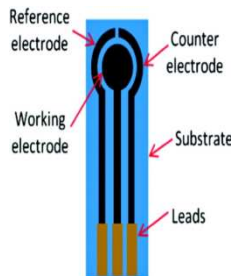
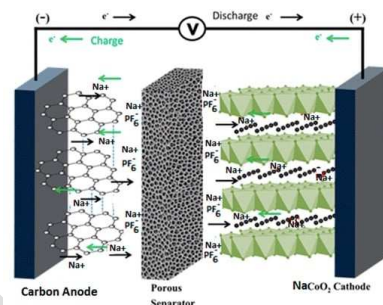




Department of Chemistry
Model Question Paper-2 with solution (2022) scheme

Module-1		
1. a	Explain the working principle of Potentiometric sensors and Thermal sensors (Flame photometer) Thermometric Sensors: It is based on the measurement of thermal changes during the interaction between analyte and receptor. Working: Main component is a small tubular catalytic reactor fitted with a temperature transducer. Analyte is fed into the reactor. The wall of the reactor is coated with a catalyst which is capable of catalyzing the reaction, liberating the heat energy. Heat liberated is quantified by transducer and convert into voltage and fed to the data storage and processing unit. The two main transducers which convert change in temperature into an electric signal are Resistive transducers: Most commonly used resistive transducer is the thermistor. It is a semiconductor device made up of oxides of transition metals. Thermocouple: It is a device which converts the temperature difference into an electrical voltage. Applications: Us Used in determination of metabolites, bioprocess monitoring and environmental control and determination of combustible gases.	7M
b	Write a note on Disposable Sensors? Explain its advantages over classical sensors. Disposable sensors: Disposable sensors are low-cost and easy-to-use sensing devices designed for short-term or rapid single-point measurements. The disposable sensors are a type of paper over which receptor and electrodes are coated. Reference electrode, working, counter electrode and receptor is printed on a single platform as shown in the fig. These electrodes are called screen-printed electrodes and are main components of disposable sensors. Advantages of disposable sensor over classical sensors ➤ They transduce physical, chemical, or biological changes in their environment to an analytical signal. ➤ Disposable sensors are biodegradable and sustainable ➤ They have a short duration of analysis and fast response times. ➤ It provides digitized chemical and biological information. Prevents the contamination of sample. 	7M
c	Describe the construction, working and applications of Sodium-ion batteries and mention any four applications.	6M

	Composition of the battery: Reactive species at anode : Carbon Reactive species at cathode : NaCoO₂ Electrolyte : Ethylene carbonates (EC), Diethyl carbonate (DEC) Separator : Polypropylene Output Voltage : 1.85 to 3.45 V Working of SIB: During charging sodium ions in cathodic side (positive electrode) is migrated and move towards anodic side (negative electrode) <i>Cathodic Reaction:</i> $\text{NaCoO}_2 \rightarrow \text{Na}_{(1-x)}\text{CoO}_2 + x\text{Na}^+ + xe^-$ <i>Anodic Reaction:</i> $x\text{Na}^+ + xe^- + 6\text{C} \rightarrow x\text{NaC}_6$ <i>Overall Reaction:</i> $\text{NaCoO}_2 + 6\text{C} \leftrightarrow \text{Na}_{(1-x)}\text{CoO}_2 + x\text{NaC}_6$ During discharging sodium ions move from anode to cathode. <i>Anodic Reaction:</i> $x\text{NaC}_6 \rightarrow x\text{Na}^+ + xe^- + 6\text{C}$ <i>Cathodic Reaction:</i> $\text{Na}_{(1-x)}\text{CoO}_2 + x\text{Na}^+ + xe^- \rightarrow \text{NaCoO}_2$ <i>Overall Reaction:</i> $\text{Na}_{(1-x)}\text{CoO}_2 + x\text{NaC}_6 \rightarrow \text{NaCoO}_2 + 6\text{C}$ Advantages of SIB: ➤ Rechargeable sodium ion for energy storage. ➤ Easier to recycle ➤ Low market prices ➤ Capable of working at room temperature, good efficiency. Disadvantages of SIB: ➤ Large ionic size Na⁺ which require more power to keep energy flowing. ➤ It takes seven days to charge in case you forget to charge it. ➤ Lower operating voltage. ➤ Need high temperature for optimal work Applications of SIB: The Sodium- ion batteries are used in mobile phones, cameras, calculators, LCD TVs, pagers, to operate laptop computers and in aerospace applications.	
2.a	Explain the working principle of electrochemical sensors, and mention its applications Electrochemical Sensors: These sensors use electrode as transducer component. The main components of electrochemical sensors are working or sensing electrode, electrolyte, counter and reference electrode. Electrolyte is a part of the electrochemical Sensors and role is to transport charge within the sensors, contact all electrode effectively, it is stable under all conditions. Following steps are involved in working of an electrochemical sensor 1. Diffusion of the analyte to the electrode/electrolyte interface (in the liquid form) 2. Adsorption onto the electrode surface 3. Electrochemical reaction with electron transfer	6M



	4.Desorption of the product 5. Diffusion of the products away from the reaction zone to the bulk of electrolyte or gas phase. Applications: 1. The oxygen sensors are used to determine dissolved oxygen in boiler water and to monitor dissolved oxygen concentrations in hydrogen fuel cell. 2. Used in security and defense applications like detection of toxic gases. 3. Used in water analysis and environmental monitoring. 4. Used in diagnostic and health care applications. 5. Used in soil parameter analysis and in agricultural applications	
b	What are Actuators & Transducers? Explain about detection of Glyphosate with electrochemical oxidation. Actuators: An actuator is a device that produces a motion by converting energy and signals going into the system. Transducers: Transducers is used to convert the signal created by the receptor-analyte interaction into a readable value. Electrochemical Sensor for Glyphosate Detection The sensor is a silicon- based chip comprising of three-electrode system. It is fabricated by electro deposition technique. Working Electrode: A gold electrode of 4 mm diameter coated with 200nm thickness gold nanoparticles Counter electrode: A gold electrode of 4 mm diameter coated with 20nm thickness gold nanoparticles Reference Electrode: Ag/AgCl/Cl Electrolytes are added to increase the conductivity of the solution and minimizes the resistance between the working and counter electrode. Working: The electrochemical detection is based on the oxidation of Glyphosate on gold working electrode. A potential of 0.78V is applied on working electrode, there is a interaction between analyte and electrode surface. Glyphosate oxidizes on the working electrode brings a change in current in the electrolyte medium. The change in the current is a measure of concentration of Glyphosate $\text{HO-P(=O)(OH)-CH}_2\text{-NH-CH}_2\text{-COOH} \xrightarrow{(1)} \text{H}_3\text{C-NH-CH}_2\text{-COOH}$ $\text{HO-P(=O)(OH)-CH}_2\text{-NH-CH}_2\text{-COOH} \xrightarrow{(2)} \text{H}_2\text{N-CH}_2\text{-P(=O)(OH)}_2$ $\text{H}_3\text{C-NH-CH}_2\text{-COOH} \rightarrow 3 \text{CO}_2 + \text{NO}_3^- + \text{PO}_4^{3-} + 24 \text{H}^+ + 20 \text{e}^-$ $\text{H}_2\text{N-CH}_2\text{-P(=O)(OH)}_2 \rightarrow 3 \text{CO}_2 + \text{NO}_3^- + \text{PO}_4^{3-} + 24 \text{H}^+ + 20 \text{e}^-$	7M
c	What are batteries? Explain the working Principle, Properties andApplications of Quantum Dot sensitized solar cells.	7M

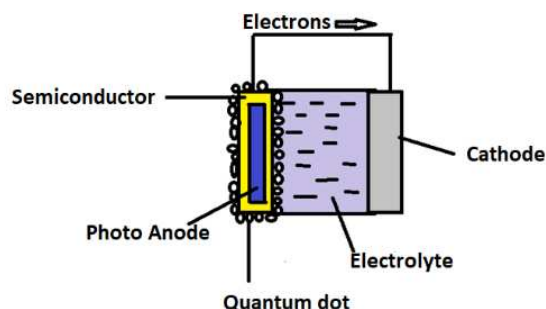
A battery is an electrochemical cell, it consists of one or more galvanic cells connected in series or parallel connection. It converts chemical energy into electrical energy during discharging and charging.

Principle

- Quantum dots are considered to be artificial atoms.
- Quantum dots (QDs) are semiconductor particles a few nanometres in size, having optical and electronic properties that differ from those of larger particles.
- A quantum dot solar cell (QDSC) is a solar cell that uses quantum dots as the absorbing photovoltaic material.
- It is used to replace bulky materials such as silicon, or copper indium gallium selenide.
- Quantum dots have band gaps that are adjustable through a wide array of energy levels by changing the size of the dots.

Construction:

- Photo Anode: It is conducting glass over which semiconductor is coated (TiO₂). Outer layer of photoanode is coated with quantum dots (QDs).
- Electrolyte: Photo anode is contact with redox electrolyte. It is hole conductor. Polysulphide is used as electrolyte.
- Cathode Electrode: It is used to regenerate electrolyte and complete the circuit.

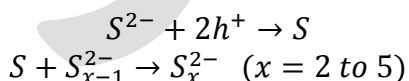


Working:

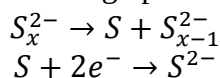
QDs are exposed to sunlight.

QDs absorb solar energy, electrons move from valence band to conduction band. These electrons are transferred to semiconductor, leaving behind holes on the surface of QD's

1. Electrolyte take up the holes from the surface of QD's and get reduced.

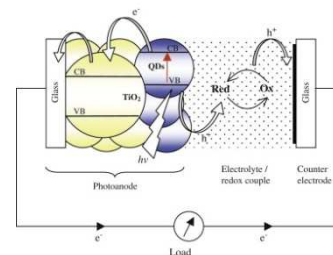


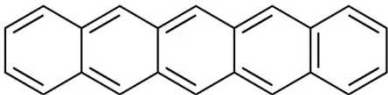
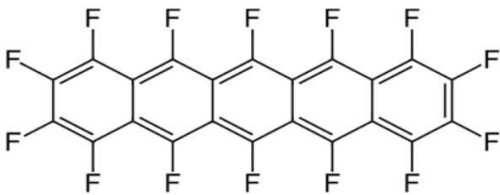
2. Electrons flow from anode to cathode through external circuit.
3. At cathode, electrolyte is regenerated taking up electrons from cathode.



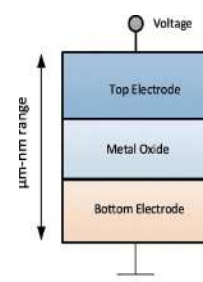
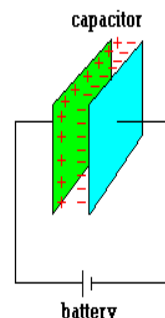
Applications of QDSSC:

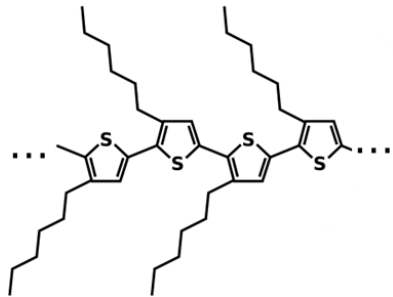
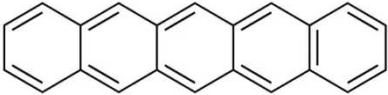
- Used for biological labeling.
- Imaging and detection and as efficient fluorescence resonance energy transfer

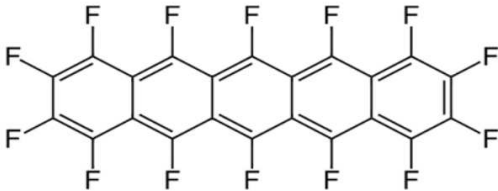


	donors. ➤ It is used as light-emitting diodes, photoconductors, photodectors and photovoltaic. ➤ It is used in biomedicine and environment. ➤ It is used in catalysis and other reactions.	
Module-2		
3.a	Explain the types of organic memory devices by taking p-type and n-type semiconductor materials. The p-Type Organic Semiconductor Material “Pentacene” An Organic molecule with π conjugated system and possesses holes as major charge carrier is called p-type semiconductor. Ex: Pentacene These molecules show bitable states when external field is applied i.e. ON and OFF state. It is linearly fused aromatic compound with five benzene rings. It can be obtained in crystal and thin film form. It shows good hole mobility, hence it behaves as a p-type semiconductor.  Pentacene The n-type organic semiconducting material Perfluoropentacene When all the hydrogen atom of Pentacene is replaced by Fluorine atoms, it formed Perfluoropentacene. Basically Fluorine is electron withdrawing nature. Hence it converts these molecules into n-type semiconductor.  Perfluoropentacene	07

b	What are Memory Devices? Explain the Classification of electronic memory devices with examples. Memory Devices: Electronic memory devices can be divided into 4 types depending the type of material it is made of. 1. Transistor-Type Electronic Memory Devices 2. Capacitor-Type Electronic Memory Devices 3. Resistor-Type Electronic Memory Devices 4. Charge Transfer Type Electronic Memory Devices Transistor-Type Electronic Memory Transistors are made from silicon, a semiconductor. It is converted to p-type and n-type semiconductor by doping trivalent and pentavalent impurities. Transistors are made using p-type and n type semiconductor. A transistor is a miniature electronic component that can work either as an amplifier or a switch. A computer memory chip consists of billions of transistors; each transistor is working as a switch, which can be switched ON or OFF .Each transistor can be in two different states and store two different numbers, ZERO and ONE. Since chip is made of billions of such transistors and can store billions of Zeros and ones, and almost every number and letter can be stored. Capacitor-Type Electronic Memory A capacitor consists of two metal plates which are capable of storing an electric charge. It is used to store data. It is like a battery that holds data based on energy. If the capacitor is charged, it holds the binary numeral, “1” and holds “0” when the cell is discharged. If the parallel plates of a capacitor are separated by dielectric layer, charges dissipate slowly and memory would be volatile. On the other hand, if the medium between the electrodes is ferroelectric in nature, can maintain permanent electric polarization that can be repeatedly switched between two stable states (bistable) by an external electric field. Thus, memory based on ferroelectric capacitors (FeRAM) is non-volatile memory. Resistor-Type Electronic Memory Memory devices containing switchable resistive materials are classified as resistor-type memory, or resistive and random access memory (RRAM). Resistor-type electronic memory usually has a simple structure, having a metal-insulator-metal structure generally referred to as MIM structure. The structure comprises of an insulating layer (I) sandwiched between the two metal (M) electrodes and supported on a substrate (glass, silicon wafer, plastic or metal foil). Initially, the device is under high resistance state or “OFF” and logically “0” state, when resistance changed or under external applied field changes to lower resistance state or “ON” logical value “1”. Charge Transfer Effects Type Electronic Memory A charge transfer (CT) complex is defined as an electron donor– acceptor (D–A) complex, characterized by an electronic transition to an excited state in which a partial transfer of charge occurs from the donor moiety to the acceptor moiety. The conductivity of a CT complex is dependent on the ionic binding between the D–A components.	06
c	What are nanomaterials? Explain any four properties of Polythiophene (P3HT)	07



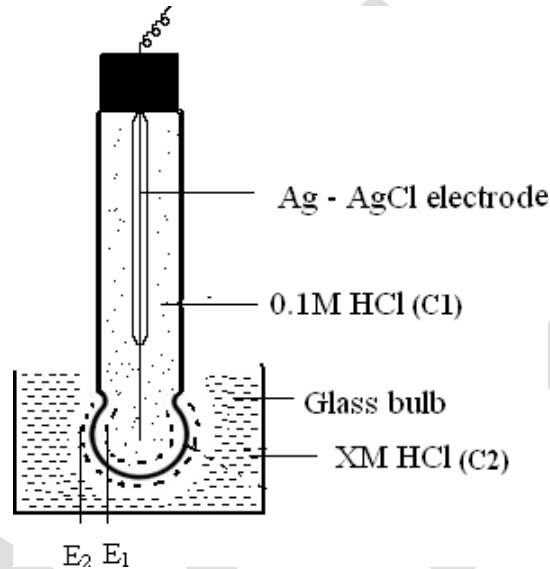
	suitable for optoelectronic devices. Nanomaterials (Silicon Nanocrystals) for Optoelectronic devices Any substance in which at least one dimensions is less than 100nm is called nanomaterials. Organic materials for Optoelectronic devices [Light absorbing materials – Polythiophene (P3HT)] Polythiophene are an important class of conjugated polymers, environmentally and thermally stable material. Chemical structure of P3HT Poly (3-hexylthiophene) is a polymer with chemical formula $(C_{10}H_{14}S)_n$. It is a Polythiophene with a short alkyl group on each repeat unit. Highly ordered (P3HT) are composed of closely packed, p-p stacked (p-p distance of 0.33nm) fully extended chains which are oriented perpendicular to the substrate. Structure of P3HT Properties: 1. P3HT is a semiconducting polymer with high stability and exhibits conductivity due to holes therefore considered as p-type semiconductor. 2. Poly-3-hexylthiophene (P3HT) has great capability as light-absorbing materials in organic electronic devices. 3. P3HT has a crystalline structure and good charge-transport properties required for Optoelectronics. 4. P3HT has a direct-allowed optical transition with a fundamental energy gap of 2.14eV. 5. Fundamental band gap of P3HT is 490nm visible region, corresponding to $\pi \rightarrow \pi^*$ transition, giving electron-hole pair. P3HT indicates that an increase in the conductivity is associated with an increase in the degree of crystallinity. 	
	OR	
4.a	Explain the types of organic memory devices by taking p-type and n-type semiconductor materials The p-Type Organic Semiconductor Material “Pentacene” An Organic molecule with π conjugated system and possesses holes as major charge carrier is called p-type semiconductor. Ex: Pentacene These molecules show bistable states when external field is applied i.e. ON and OFF state. It is linearly fused aromatic compound with five benzene rings. It can be obtained in crystal and thin film form. It shows good hole mobility, hence it behaves as a p-type semiconductor.  Pentacene The n-type organic semiconducting material Perfluoropentacene	06

	When all the hydrogen atom of pentacene is replaced by Fluorine atoms, it formed Perfluoropentacene. Basically Fluorine is electron withdrawing nature. Hence it converts this molecules into n-type semiconductor.  Perfluoropentacene	
b	Mention any four properties and applications of LC-displays <u>Properties</u> ➤ Liquid crystal can flow like a liquid, due to loss of positional order ➤ These are elongated and have some degree of rigidity ➤ They have less orientational order ➤ Transition from crystalline solids to liquid crystals caused by a change of temperature. <u>Applications of liquid crystals:</u> 1. Liquid crystals are used in watches, calculators, mobile telephones, laptops, computers etc. 2. These are used in blood pressure instrument, digital thermometers and TV Chanel indicators. 3. These are used in potentiometer, conductometer. 4. used in computer monitors	07
c	Mention any four properties and applications of OLED Organic Light Emitting Diodes (OLED's) OLEDs are thin film devices consisting of a stack of organic layers sandwiched between two electrodes. OLEDs operate by converting electrical current in to light via an organic emitter. OLED is an electroluminescent device that uses organic molecules as a source of light emission. Light is emitted by organic material when an external field is applied across it. <u>Properties:</u> Some of the key properties of Organic Light Emitting Diodes (OLEDs) include: 1. Thinness and flexibility: OLEDs are very thin and flexible, which makes them suitable for use in curved or flexible displays. 2. High contrast: OLEDs have a high contrast ratio, which means that they can produce deep blacks and bright whites, resulting in images with vivid and rich colours. 3. Fast response time: OLEDs have a fast response time, which means that they can switch on and off quickly, resulting in smooth and seamless motion in video content.	07

	4. Wide viewing angle: OLEDs have a wide viewing angle, which means that the image quality is maintained even when viewed from different angles. 5. Energy efficiency: OLEDs are energy efficient as they do not require a back light like traditional LCD displays, resulting in lower power consumption. Applications 1. Flat-panel TV screen 2. Digital cameras Mobile phones	
	Module-3	
5.a	Define corrosion? Mention at least six implications of corrosion “The spontaneous deterioration and consequent loss of a metal/alloy due to chemical/electrochemical attack by the environment”.	07
b	Explain: (i) Differential metal corrosion & (ii) Water-line corrosion <u>Differential metal corrosion (Galvanic corrosion)</u> 1) When two dissimilar metals are in direct contact with one another and exposed to a corrosive medium, the metal with lower electrode potential becomes anode and suffers from corrosion, whereas the metal with higher electrode potential becomes cathode and protected from corrosion. This type of corrosion is known as differential metal corrosion or galvanic corrosion. 2) The rate of corrosion depends mainly on the difference in the position of the two metals in galvanic series. 3) Higher the difference, faster is the rate of corrosion. Example: Electrode potential of iron (-0.44 V) is less than that of copper (0.34 V). Therefore when iron is in contact with copper, iron becomes anodic and undergoes corrosion, whereas copper becomes cathodic and protected from the corrosion. At Anode: $Fe \rightarrow Fe^{2+} + 2e^{-}$ At Cathode : $H_2O + \frac{1}{2}O_2 + 2e^{-} \rightarrow 2OH^{-}$ $Fe^{2+} + 2OH^{-} \rightarrow Fe(OH)_2$ $2Fe(OH)_2 + \frac{1}{2}O_2 \rightarrow Fe_2O_3 \cdot 2H_2O$	06
c	Explain the construction and working of glass electrode Glass Electrode: This electrode works on the principle that when a thin, low resistivity glass membrane is in contact with a solution containing H^{+} ions, a potential develops across the membrane and the solution. Potential developed depends on the concentration of hydrogen ions in the solution.	06

Construction:

- It consists of a long glass tube with a thin-walled glass bulb containing 0.1 M HCl [C_1]. Ag/AgCl electrode placed inside the solution to provide electrical contact.
- The glass electrode is dipped in unknown solution of concentration C_2 .
- The potential developed across the membrane by the exchange of ions with the composition of glass is known as the boundary potential E_b ($E_2 - E_1$).
- Even when $C_1 = C_2$, $E \neq 0$ a small potential is developed across the membrane, it is called as asymmetric potential (E_{asy}).



Cell representation: Ag / AgCl / HCl (0.1M) / unknown solution / Glass

Working:

E_1 & E_2 is the potential developed at outer and inner membrane respectively the boundary potential is, $E_b = E_2 - E_1$

$$E_b = E^0 + \frac{0.0591}{n} \log[C_2] - \left\{ E^0 + \frac{0.0591}{n} \log[C_1] \right\}$$

$$E_b = \frac{0.0591}{n} \log[C_2] - \frac{0.0591}{n} \log[C_1]$$

$$E_b = \frac{0.0591}{n} \log[C_2] + \frac{0.0591}{n} \log \frac{1}{[C_1]}$$

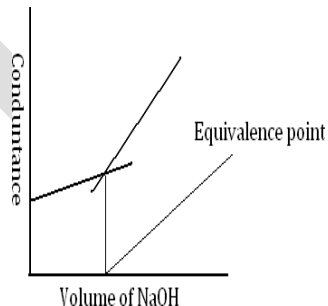
$$\text{Where } L = \frac{0.0591}{n} \log \frac{1}{[C_1]}$$

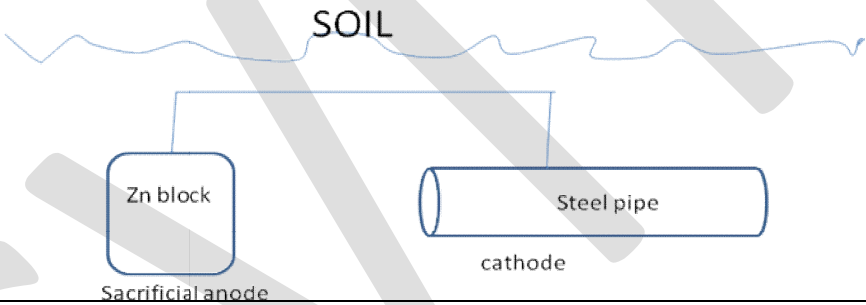
Therefore

$$E_b = L + \frac{0.0591}{n} \log[C_2]$$

Here $n=1$, $[C_2] = [H^+]$ ions in outer membrane, Then

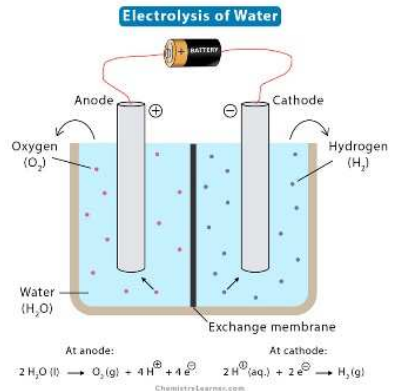
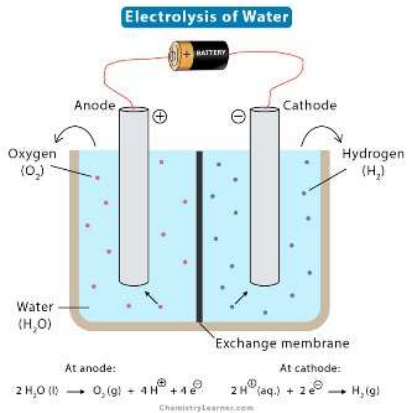
$$E_b = L + 0.0591 \log [H^+]$$

	$E_b = L - 0.0591p^H \quad (pH = -\log[H^+])$ The Glass electrode potential is given by $E_G = E_b + E_{Ag/AgCl} + E_{Asy}$ Substitute the value of E_b $E_G = L - 0.0591p^H + E_{Ag/AgCl} + E_{Asy}$ Consider $L_1 = L + E_{Ag/AgCl} + E_{Asy}$, Therefore $E_G = L_1 - 0.0591p^H$ OR	
6.a	Explain the application of conductometric electrode in estimation of weak acid. Weak acid v/s Strong base Consider the titration of acetic acid against NaOH. The conductance of the acid will be initially low since acetic acid is a weak electrolyte. When NaOH is added to the acid, the salt formed is highly ionized and the conductance increases. On complete neutralization of the acid, further addition of base leads to an increase in the number of mobile OH^- ions. Hence the conductance increases sharply. $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$ A plot of conductance against the volume of base added is shown in the figure. The point of intersection of two curves gives the neutralization point. 	06
6.b	Explain: i) corrosion control by Anodization & ii) Sacrificial anodic Method. Anodizing Anodizing is the process of oxidation of outer layer of metal to its metal oxide by electrolysis. Oxide layer formed over the metal itself acts as protective layer. Anode : Aluminium Cathode : Steel or Copper Electrolyte : 5-10% chromic acid Temperature : 30-40 °C Applied potential : 0-40V Aluminum is made as anode and copper or steel is used as cathode. The electrodes are immersed in an electrolyte consisting of 5-10% chromic acid. The temperature is maintained at 30-40 °C. Potential of 0-40 V is applied which oxidizes outer layer of Al to Al_2O_3 and that gets deposited over the metal. The reactions involved are as follows	07

	<i>At Anode:</i> $2Al(s) + 3H_2O(l) \rightarrow Al_2O_3(s) + 6H^+ + 6e^-$ <i>At Cathode:</i> $6H^+ + 6e^- \rightarrow 3H_2(g)$ <i>Over all reaction:</i> $2Al(s) + 3H_2O(l) \rightarrow Al_2O_3(s) + 3H_2(g)$ The anodized coatings are thicker than the natural oxide film and possess improved corrosion resistance as well as resistance to mechanical injury. Sacrificial anodic protection: 1) The metal to be protected is connected to a more anodic metal using a metallic wire. 2) The more active metal gets corroded while the parent structure is protected from corrosion. 3) The more active metal so employed is called sacrificial anode. 4) The sacrificial anodes to be replaced by fresh ones as and when it is required. 5) Commonly used sacrificial anodes are: Mg, Zn, Al etc. Example: steel pipe is protected by connecting it to a block of Zn. In such cases steel acts as a cathode and is unaffected or protected from corrosion and Zn act as anode and undergoes sacrificial corrosion. 																
6.c	What is CPR? A thick brass sheet of area 100 inch² is exposed to moist air. After 1 years of period, it was found to experience a weight loss 75 g due to corrosion. If the density of brass is 2.52 g/cm³. Calculate CPR in mpy and mmpy. Corrosion Penetration Rate (CPR) The corrosion penetration rate (CPR) is defined as “the speed at which any metal in a specific environment deteriorates due to chemical reactions when it is exposed to a corrosive environment” or the amount of corrosion loss per unit time in thickness or the speed at which corrosion spreads to the inner portions of a material. To calculate CPR in mpy <table border="1"> <thead> <tr> <th></th><th>Given</th><th>CPR in mpy</th></tr> </thead> <tbody> <tr> <td>K</td><td></td><td>534</td></tr> <tr> <td>W (wt loss)</td><td>75 g</td><td>75 x1000 mg</td></tr> <tr> <td>P</td><td>2.52g/cm³</td><td>2.52g/cm³</td></tr> <tr> <td>A</td><td>100 inch²</td><td>100 inch²</td></tr> </tbody> </table>		Given	CPR in mpy	K		534	W (wt loss)	75 g	75 x1000 mg	P	2.52g/cm ³	2.52g/cm ³	A	100 inch ²	100 inch ²	06
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A	100 inch ²	100 inch ²															

	<table> <tr> <td>T</td><td>1 years</td><td>1x365x24 hrs</td></tr> </table> $CPR = \frac{k \times W}{D \times A \times T}$ $CPR = \frac{534 \times 75}{2.52 \times 100 \times 1 \times 365 \times 24}$ $CPR = 0.0181 \text{ mpy}$ To calculate CPR in mmpy <table> <tr> <td></td><td>Given</td><td>CPR in mmpy</td></tr> <tr> <td>K</td><td></td><td>87.6</td></tr> <tr> <td>W (wt loss)</td><td>75 g</td><td>75 x 1000 mg</td></tr> <tr> <td>ρ</td><td>2.52 g/cm³</td><td>2.52 g/cm³</td></tr> <tr> <td>A</td><td>100 inch²</td><td>100 inch²</td></tr> <tr> <td>t</td><td>1 years</td><td>1x365x24 hrs</td></tr> </table> $1 \text{ inch}^2 = 6.45 \text{ cm}^2$ $1 \text{ cm}^2 = 0.155 \text{ inch}^2$ $CPR = \frac{87.6 \times W}{D \times A \times T}$ $CPR = \frac{87.6 \times 75}{2.52 \times 100 \times 6.45 \times 1 \times 365 \times 24}$ $CPR = 4.061 \times 10^{-3} \text{ mmpy}$	T	1 years	1x365x24 hrs		Given	CPR in mmpy	K		87.6	W (wt loss)	75 g	75 x 1000 mg	ρ	2.52 g/cm ³	2.52 g/cm ³	A	100 inch ²	100 inch ²	t	1 years	1x365x24 hrs	
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ρ	2.52 g/cm ³	2.52 g/cm ³																					
A	100 inch ²	100 inch ²																					
t	1 years	1x365x24 hrs																					
	Module4																						
7.a	A polydisperse sample of polystyrene is prepared by mixing three monodisperse samples in the following proportions. 1g of 10000 molecular weight, 2g of 50000 molecular weight and 2g of 100000 molecular weight. Determine number average and weight average molecular weight. Find the index of polydispersity. <u>Number average molecular mass:</u> $\overline{M}_n = \frac{n_1 M_1 + n_2 M_2 + n_3 M_3}{n_1 + n_2 + n_3}$ $\overline{M}_n = \frac{\sum N_i M_i}{\sum N_i}$	07																					

	$= \frac{N_1 M_1 + N_2 M_2 + N_3 M_3}{N_1 + N_2 + N_3 \dots}$ $= \frac{1 \times 10000 + 2 \times 50000 + 2 \times 100000}{1 + 2 + 2}$ $= 62000 \text{ g/mol}$ $\overline{M_w} = \frac{N_1 M_1^2 + N_2 M_2^2 + N_3 M_3^2}{N_1 M_1 + N_2 M_2 + N_3 M_3}$ $\overline{M_w} = \frac{1 \times (10^4)^2 + 2 \times 5(10^4)^2 + 2 \times (10^5)^2}{1 \times 10000 + 2 \times 50000 + 2 \times 100000} = 670967.74 \text{ g/mol}$ Poly dispersity index, $\text{PDI} = \frac{\overline{M_w}}{\overline{M_n}} = \frac{670967.74}{62000} = 10.82$	
7.b	Explain the Preparation, properties, and commercial applications of Kevlar. <u>Kevlar:</u> Kevlar is synthesized in solution of N-methyl-pyrrolidone & calcium chloride from the monomers 1, 4-phenylene-diamine (para-phenylenediamine) & terephthaloyl chloride through a condensation reaction with liberation of HCl as a byproduct. 1,4-Phenylene-diamine + Terephthaloyl chloride → Kevlar <u>Properties:</u> 1. It has very light weight. 2. It has high tensile strength and stiffness. 3. It has very good corrosion resistance. <u>Applications of Kevlar:</u> 1. It is used to make light weight boat hulls 2. aircraft fuselage panels, pressure vassals, high performance race car, bullet proof vests, Used in puncture resistance bicycle tyres etc.	07
7.c	Describe the hydrogen production by photo catalytic water splitting method. Generation of hydrogen by electrolysis of water It is the process of splitting of water into oxygen and hydrogen gas by electrolysis. ➤ It consists of two electrodes i.e. anode and cathode.	06

	➤ Both are separated by membranes. ➤ When electricity is passed, oxidation takes place at anode, it gives H^+ ions and electron, also liberates Oxygen gas. At Anode: $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$ ➤ The H^+ ions move into cathodic compartment through membranes and electrons move from anode to cathode through external circuit. ➤ At cathode the H^+ ions accepts electrons and forms H_2 gas. This liberated hydrogen gas is used as a fuel ➤ At Cathode : $4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2(\text{g})$		
	OR		
8.a	Describe the hydrogen production by photo catalytic water splitting method. It is the process of <u>splitting of water</u> into <u>oxygen</u> and <u>hydrogen</u> gas by <u>electrolysis</u>. ➤ It consists of two electrodes i.e. anode and cathode. ➤ Both are separated by membranes. ➤ When electricity is passed, oxidation takes place at anode, it gives H^+ ions and electron, also liberates Oxygen gas. At Anode: $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$ ➤ The H^+ ions move into cathodic compartment through membranes and electrons move from anode to cathode through external circuit. ➤ At cathode the H^+ ions accepts electrons and forms H_2 gas. This liberated hydrogen gas is used as a fuel At Cathode : $4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2(\text{g})$		07
b	Mention the properties of hydrogen pertaining to fuel and its advantages in production of energy properties of hydrogen pertaining to fuel and its advantage It is lightest element Gas of diatomic molecules having the formula H_2. Colourless, odourless and Tasteless Non-metallic and non-toxic Highly combustible,		06

	Advantages Eco friendly in nature High efficient energy output It's a Renewable Energy Source and Bountiful in Supply It is Practically a Clean Energy Source Used for powering space ships	
c	What are green fuels? Explain the advantages & disadvantages of photovoltaic cells. Definition of Green fuels: A fuel derived from biomass is called as green fuel. Advantages of PV cells: 1. Eco friendly energy conversion device. 2. Do not undergo corrode 3. No moving parts in PV cell, hence no wear and tear. Disadvantages of PV cells: 1. Installation cost is high. 2. Produced only during the day time. 3. It generates only DC current.	06
	Module-5	
9.a	What are e-waste and explain the need for e-waste management <u>Definition of waste.</u> Waste refers to any material that is not wanted or needed and is discarded or disposed of. <u>Need of E- Waste Management:</u> E-waste management is necessary because it helps to address several environmental, health, and resource-related concerns, including: ➤ Protecting the environment: E-waste contains toxic substances, such as lead, mercury, and cadmium that can have harmful effects on the environment and human health if not properly managed. ➤ Conserving resources: E-waste contains valuable resources, such as metals, that can be recovered and reused through proper recycling. ➤ Reducing greenhouse gas emissions: The production of new electronic products releases greenhouse gases, such as carbon dioxide, into the atmosphere. Proper recycling and disposal of e-waste can reduce the environmental impact of electronic products. ➤ Reducing land filling: land filling of electronic waste can result in the release of toxic materials into the environment and contribute to soil and water pollution. ➤ Protecting public health: Improper handling and disposal of e-waste can expose workers and the general public to hazardous materials and cause serious health problems. ➤ E-waste can be toxic, is not biodegradable and accumulates in the environment, in the soil, air, water and living things. ➤ Causes air pollution, Soil pollution and water pollution. Electronic Recycling Promotes Soil Fertility and Maintain Nutrient.	07
b	Explain the health hazard due to exposure to e-waste.	07

	<u>Health hazardous due to exposure of e waste:</u> Exposure to electronic waste (e-waste) can have serious health consequences, including: Poisoning: E-waste can contain toxic substances, such as lead, cadmium, and mercury that can cause poisoning if they enter the body. Respiratory problems: Exposure to dust and fumes generated during the dismantling and disposal of e-waste can cause respiratory problems, such as asthma and bronchitis. Neurological effects: Exposure to toxic substances in e-waste, such as lead and mercury, can cause neurological effects, including memory loss, tremors and coordination problems. Reproductive problems: Exposure to toxic substances in e-waste, such as cadmium, can cause reproductive problems, including infertility and birth defects. Cancer: Exposure to carcinogenic substances, such as dioxins and polychlorinated biphenyls (PCBs), found in e-waste, can increase the risk of cancer It is important to effectively manage e-waste to minimize the exposure of workers and the public to hazardous materials and to prevent serious health problems.	
c	Role of s stakeholders - producers, consumers, recyclers, and statutory bodies. In the management of electronic waste (e-waste), the following stakeholders play important unique role in the management of e-waste: 1. Producers - are responsible for designing and producing electronic products and may also be involved in the collection and recycling of e- waste. 2. Consumers - play a crucial role in the responsible disposal of e-waste and making informed choices about the purchase of electronic products. 3. Recyclers - are responsible for safely and responsibly managing e-waste, including the collection, dismantling, and recycling of electronic waste. 4. Statutory bodies - such as governments, are responsible for creating and enforcing regulations and policies to manage e-waste and promoting public awareness and education about e-waste management.	06
	OR	
10. a	Toxic materials used in manufacturing Electronic and Electrical products. Electronic and electrical products can contain a variety of toxic materials,including: 1. Lead: Lead is a toxic heavy metal commonly used in the manufacture of batteries, computer monitors, and other electronic components. 2. Mercury: Mercury is used in some fluorescent lights, batteries, and other electronic devices. 3. Cadmium: Cadmium is a toxic heavy metal used in rechargeable batteries, pigments, and plastic stabilizers. 4. Polyvinyl Chloride (PVC): PVC is a common plastic used in electronic cables and	07

	other components. It can release toxic chemicals, such as dioxins, when burned or during disposal. 5. Brominated flame retardants (BFRs): BFRs are used in the manufacture of electronic products to prevent fires. However, they are toxic and can harm the environment and human health. 6. Barium: Barium is used in some electronic components, including cathode ray tubes. 7. Rechargeable Batteries contains Lithium is used in batteries, but it can be toxic if not handled properly. Cadmium, Lead, Sodium, Lithium, Nickel etc., Chlorofluorocarbons (CFCs) are toxic chemicals that were widely used as coolants and solvents in electronic products, such as refrigerators, air conditioners. They cause ozone depletion.	
b	Explain the advantages of recycling and recovery in e-wastes. <u>Advantages of recycling</u> 1. Conserving natural resources. 2. Protecting ecosystems and wildlife. 3. Reducing demand for raw materials. 4. Saving energy. 5. Cutting climate-changing carbon emissions. 6. Cheaper than waste collection and disposal <u>Advantages of recovery</u> Recycling recovers valuable materials from old electronics that can be used to make new products. As a result, we save energy, reduce pollution, reduce greenhouse gas emissions and save natural resources by extracting fewer raw materials from the earth.	06
c.	Explain about sources, composition and characteristics of e-waste. <u>Sources of electronic waste (e-waste):</u> The main sources of electronic waste (e-waste) include: 1. Consumer electronics such as smart phones, laptops, televisions, and household appliances. 2. Office equipment such as computers, printers, copiers, and fax machines. 1. Medical equipment such as X-ray machines, monitors, and diagnostic equipment. 2. Electronic toys and games. 3. Military and government surplus equipment. 4. Obsolete technology such as outdated computer equipment and VCRs. 5. Discarded or broken electronic devices. 6. Colleges often use electronic equipment such as projectors, computer equipment, and audio-visual equipment in classrooms and laboratories. <u>Composition of E-waste</u>	07

	The composition of electronic waste (e-waste) can vary depending on the type of device and its components. In general, e-waste contains a mixture of materials including: 1. Metals: E-waste often contains valuable metals such as copper, gold, silver, and aluminium. 2. Plastics: Many electronic devices contain plastic components, including casings, insulation, and cables. 3. Glass: Electronic devices often contain glass components, such as screens and lenses. 4. Circuit boards: Many electronic devices contain circuit boards, which contain a mixture of metals and other materials. 5. Batteries: Some electronic devices contain batteries, which can contain hazardous materials such as lead, mercury, and cadmium. 6. Other hazardous materials: E-waste may also contain other hazardous materials, such as flame retardants, heavy metals, and polychlorinated biphenyls (PCBs). <u>Characteristics of E-waste:</u> The characteristics of electronic waste (e-waste) can include: 1. Complexity: E-waste often contains a complex mixture of materials, making it challenging to recycle and dispose of properly. 2. Hazardousness: E-waste can contain hazardous materials such as heavy metals, flame retardants, and batteries, which can pose significant environmental and health risks. 3. Volatility: E-waste is a rapidly growing waste stream due to the increasing use of electronic devices and the limited lifespan of many electronic products. 4. Global issue: E-waste is a global issue, as electronic devices are manufactured, used, and discarded worldwide. 5. Resource depletion: The extraction of raw materials for electronic devices contributes to resource depletion, and the improper disposal of e-waste can lead to environmental contamination and waste of valuable resources. 6. Environmental and health impacts: Improper disposal of e-waste can lead to significant environmental and health impacts, including soil and water contamination, air pollution, and harm to human health.	
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